Prefix Memory and the Growth Spectrum of Event Observers

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This paper is Paper XLIV (version 1.0) of the RIME program. It classifies the minimal prefix-memory observers of the fixed circular registered-absorption action and determines their depth-dependent growth spectrum.

Abstract

On the singleton-seed circular action generated by rotation and adjacent collapse, we study observers that retain the registration origin, the first j complete selected absorption records, and all future selected events. We classify their minimal quotient on actual reachable states and count its classes for every $j \geq 1$. At fixed depth the state count is asymptotic to $5\,2^j\sqrt{n}\,4^n/(16\sqrt{\pi})$; every positive sublinear depth has exponential base 4. At depth $j = \lfloor \theta n \rfloor$, the growth base rises strictly from 4 to 9/2 for $0 < \theta < 1/3$ and remains 9/2 thereafter. Above this threshold the prefix and complete-summary counts are asymptotically equal, with exponentially vanishing relative deficit. In the square-root critical window their ratio follows the Gaussian profile $\Phi(3u/2)$, with a uniform $O(n^{-1/2})$ error on bounded u-intervals. The result exhibits intermediate observer-state growth rates on one fixed dynamics, even though the observer need not retain the full history.

Keywords: event history; prefix memory; observer-relative state minimization; circular coalescence; exponential growth spectrum

1 Introduction

How much of an old event history must a predictive observer retain? The answer depends on the report being requested. In a circular registered absorption system, the future selected-event observer can forget the old selected chain once the current placement is known, whereas a complete-summary observer reports that chain. Their exact minimal state counts have exponential bases 4 and 9/2, respectively [2]. We investigate the observations between these two contracts by retaining a prefix of the old chain.

The retained data are complete event records with source identities, not merely counts or L/R signs. A depth-j observer also distinguishes a chain shorter than j from a saturated prefix and reads all future selected events. At linear depth $j = \lfloor \theta n \rfloor$, its growth base is

$$\lambda(\theta) = \begin{cases} \dfrac{2^{2+\theta}}{(1+\theta)^{1+\theta}(1-\theta)^{1-\theta}}, & 0 < \theta \leq 1/3, \\ 9/2, & 1/3 \leq \theta < 1. \end{cases} \tag{1.1}$$

Thus every rate in $(4, 9/2)$ occurs on the same dynamics. Above the threshold the ratio to the complete-summary count tends to one; within a depth window of order $\sqrt{n}$ around $n/3$, it has a Gaussian transition. These are cardinality statements, not equality of the underlying quotients.

The proof combines future-event separation with actual chain and placement-fibre reachability. Short and saturated prefixes have different counting parameters. A uniform truncated-binomial reduction gives (1.1), while an exact chain-length reweighting resolves the critical window.

Sequential-machine equivalence supplies the general language of right indistinguishability [5, 6, 8]. We use, rather than reprove, the actual source-addressed chain and fibre classification of [3] and the constructive future separation theorem of [2, Theorem 6.2]. The new result is a prefix-indexed family of exact quotients and its uniform growth spectrum.

2 Dynamics, Observers and Imported Structure

Fix $n \geq 3$, $Q = \mathbb{Z}/n\mathbb{Z}$, and source atoms $\Omega = \{0, \dots, n-1\}$. A placement P assigns disjoint nonempty atom packets to occupied sites, covering Ω. The singleton seed is $P_0(q) = \{q\}$. The site actions are

$$p(q) = q + 1, \qquad d(1) = 0, \quad d(q) = q \quad (q \neq 1). \tag{2.1}$$

Packets follow these maps and merge by union when their images coincide. Words are chronological. An effective fusion has complete ordered record

$$\rho = (d, 0, 1, U, V, 0, U \cup V), \tag{2.2}$$

where U, V occupied ports $0, 1$ just before the fusion.

Set $F = \{0, 1\}$. The first fusion that creates exactly F is its registration, with origin o. Once registered, a selected absorption is an effective fusion involving the current packet containing F, called the carrier. Unrelated mergers do not append a selected record. We use the full registration record (2.2) as the semantic origin o. Write $x = (P, o, A)$ for the current placement, origin and selected chain. Let X_n be the set of all such states reached from the singleton seed by finite actual words. The visible output alphabet contains null, registration outputs $\mathsf{reg}(\rho)$ and selected-absorption outputs $\mathsf{abs}(\rho)$, where ρ is a complete record (2.2). On a letter a, the output $\epsilon(x, a)$ is $\mathsf{reg}(\rho)$ when that letter first creates F, $\mathsf{abs}(\rho)$ when it fuses an already registered carrier with another packet, and null otherwise. In particular, registration itself is visible; rotations, nonfusing defects and unrelated fusions emit null. Registration has result F, whereas a selected absorption has result strictly containing F. Write $\delta_a x$ for the actual successor. The trace $\mathrm{Trace}(x, w)$ retains one tagged output, including nulls, per letter of the common raw word w.

For $j \geq 1$ define the retained label

$$L_j(x) = \begin{cases} \bot, & x \text{ unregistered}, \\ (o, \mathrm{take}_j A, \mathsf{short}), & |A| < j, \\ (o, \mathrm{take}_j A, \mathsf{saturated}), & |A| \geq j. \end{cases} \tag{2.3}$$

The short marker reports the current chain length. The registration origin is unique for this pair when $n \geq 3$.

For completeness, define the label update U_j on reachable label/output pairs. Let $\mathsf{flag}_j(k)$ be short for $k < j$ and $\mathsf{saturated}$ for $k \geq j$. Then

$$\begin{aligned} \mathsf{U}_j(z, \mathsf{null}) &= z, \\ \mathsf{U}_j(\bot, \mathsf{reg}(\rho)) &= (\rho, (), \mathsf{short}), \\ \mathsf{U}_j((o, B, \mathsf{short}), \mathsf{abs}(\rho)) &= (o, B \parallel [\rho], \mathsf{flag}_j(|B| + 1)), \\ \mathsf{U}_j((o, B, \mathsf{saturated}), \mathsf{abs}(\rho)) &= (o, B, \mathsf{saturated}). \end{aligned}$$

Here $B \parallel [\rho]$ appends one record; registration initializes an empty selected chain. The definitions give $L_j(\delta_a x) = \mathsf{U}_j(L_j(x), \epsilon(x, a))$.

Two states are equivalent for this observer if, under every common raw word, their selected-event traces and all retained-label reports agree, including the label at the empty prefix.

$$x \equiv_j y \quad \Longleftrightarrow \quad x \equiv_{\text{future}} y \ \text{ and } \ L_j(x) = L_j(y). \tag{2.4}$$

For the right-to-left implication, assume equal initial labels and equal future traces. At each letter the outputs agree, so the displayed update rule gives equal successor labels. Induction keeps the labels equal at every prefix of every common continuation. For the converse, the empty prefix gives equal current labels and agreement of all selected-event traces is exactly future equivalence. Let $Q_n^{[j]}$ count these classes on X_n, and let $Q_{n,\mathrm{reg}}^{[j]}$ count them on the successor-closed registered subset.

We record precisely the inherited structural results used below. Put $N = n - 2$.

1. **Actual chains and fibres** [3, Theorems 3.1 and 4.2]. A selected chain is uniquely encoded by nonzero checkpoints among $1, \dots, N$ and their L/R port bindings. If its final carrier adds $r < N$ atoms to F, that carrier is one of the $r + 1$ proper arcs extending F. For a proper carrier C, list its source complement in the positive linear order starting immediately after C and ending just before it. Partition that list into nonempty consecutive intervals. The blocks C and those outside intervals must occur at their occupied physical sites in the induced positive cyclic block order. For each fixed compatible complete chain, every placement satisfying these two conditions is actually reachable. The number of these placements is $n\binom{2N-r}{N-r}$, independent of the chain. At a full carrier each complete chain has n actual physical phases. These are statements about actual words, not arbitrary record lists.

2. **Future equivalence** [2, Theorem 6.1 and Proposition 3.3]. All permanently unregistrable states and registered rank-one states are silent and form one future class. Every registerable unregistered placement is its own class. Registered nonterminal states are equivalent exactly when their current placements agree. The number of registerable placements is $R_n = n\binom{2N}{N-1}$. Constructive common-suffix probes establish the separation of these classes.

The comparator is the **complete-summary Moore observer**, whose current report is $\perp$ before registration and (P, o, A) afterwards. Write s_n for the number of actual complete registered summaries, including terminal phase. Its minimal quotient count is $Q_n^{\mathrm{full}} = 1 + R_n + s_n$ [2]: registered summaries are separated by their current report, while the unregistered region has the same dead and registerable placement classes as above.

Here a checkpoint is the cumulative number $b = |C'| - 2$ of atoms absorbed beyond F, read from that record's result C'. If the preceding carrier is at port 0, the record has binding R and absorbs a prefix of the positive ordered complement; at port 1 it has binding L and absorbs a suffix. The checkpoint increment is the size of that partner. Starting at F, these increments and bindings recover the complete ordered records. When only two packets remain the possible partner sets coincide, but the two port bindings still give different records.

At depth zero, we use the future-only convention without a registration label; the theorems below concern positive depth.

3 Exact Prefix Classes

Write a, b for the atoms added to the left and right of a proper carrier, so $a + b = r < N$. Its ordered records recover each checkpoint and port binding.

Lemma 3.1.

Sum the number of compatible retained labels over all $r + 1$ carrier arcs of extension r. The result is

$$c_{0,j} = 1, \qquad c_{r,j} = 2^j \binom{r+1}{j+1} + \sum_{\ell=1}^{j-1} 2^\ell \binom{r-1}{\ell-1} \quad (r \geq 1). \tag{3.1}$$

Binomial coefficients with an index outside their range are zero.

Proof. *First count saturated prefixes for a fixed arc (a, b). If u of the j records are left absorptions, their order has $\binom{j}{u}$ possibilities. Their cumulative endpoints on the left form a u-element subset of $\{1, \dots, a\}$; the right endpoints form a $(j-u)$-element subset of $\{1, \dots, b\}$. These data recover every ordered record. Conversely every such choice is a legal prefix, which may leave some of either side for unobserved past absorptions. The fixed-arc count is*

$$\sum_{u=0}^{j} \binom{j}{u} \binom{a}{u} \binom{b}{j-u}. \tag{3.2}$$

For each u, sum over $a + b = r$. The convolution is

$$\sum_{a=0}^{r} \binom{a}{u} \binom{r-a}{j-u} = \binom{r+1}{j+1}. \tag{3.3}$$

One proof chooses a $(j+1)$-subset of $\{0, \dots, r\}$ and designates its $(u+1)$st element as a. Choosing the smaller and larger elements gives the summand. Summing $\binom{j}{u}$ yields the first term of (3.1).

For a short prefix of length $\ell < j$, the retained list is the whole current chain, so its last checkpoint is exactly r. For $r > 0$, choose $\ell - 1$ earlier checkpoints and one of two port bindings at each event. This gives $2^\ell \binom{r-1}{\ell-1}$. Its side increments determine one carrier arc; summing across arcs therefore counts it once. The empty chain occurs only at $r = 0$.

Every counted prefix is actual at every placement compatible with its fixed carrier arc. For a saturated prefix, append its remaining nonempty left and right increments in any fixed order, then apply the inherited complete-chain fibre theorem. For a short prefix, use its complete chain directly. Conversely an actual chain determines all the chosen endpoints and bindings. This proves both counting and attainability.

Lemma 3.2.

At a full carrier the number of retained labels is

$$T_{N,j} = 2^j \binom{N}{j} + \sum_{\ell=1}^{j-1} 2^\ell \binom{N-1}{\ell-1}. \tag{3.4}$$

Proof. *A saturated prefix chooses j checkpoints from $1, \dots, N$ and their j bindings. It can be completed to the full carrier if it does not already end there. A short chain must end at N, giving the second term. The actual chain theorem realizes these choices. With two packets left, absorbing the whole complement in port direction L or R produces different ordered records, so both choices remain in the count. Physical terminal phase is not part of the retained label.*

Theorem 3.3.

Define

$$A_n^{[j]} = n \sum_{r=0}^{N-1} c_{r,j} \binom{2N-r}{N-r}. \tag{3.5}$$

For every $n \geq 3$ and $j \geq 1$,

$$Q_n^{[j]} = 1 + R_n + T_{N,j} + A_n^{[j]}, \qquad Q_{n,\mathrm{reg}}^{[j]} = T_{N,j} + A_n^{[j]}. \tag{3.6}$$

The normal-form keys are: one key for permanently unregistrable states; P for registerable unregistered states; (P, L_j) for registered nonterminal states; and L_j for registered terminals.

More explicitly, let $\mathcal{R}_n$ be the actual registerable unregistered placements, $\mathcal{T}_{N,j}$ the actual terminal label image, and $\mathcal{P}(C, L)$ the actual placements compatible with a proper carrier C and retained label L. There is an equivalence

$$X_n/{\equiv_j} \;\simeq\; \{\star\} \;\sqcup\; \mathcal{R}_n \;\sqcup\; \mathcal{T}_{N,j} \;\sqcup \coprod_{C \text{ proper}} \; \coprod_{L \text{ compatible with } C} \mathcal{P}(C, L).$$

The parameter spaces use actual attainable labels and literal placements. Equality in this normal form is equivalent to the all-word observer equivalence.

Proof. *Apply (2.4) to the inherited future classes. In registered nonterminal classes, different placements remain separated by a future suffix, and different labels are separated by the current report. Equal placement and label give both equal future behaviour and equal current reports. At rank one there are no future selected events, so equality is exactly equality of labels. Registered labels differ from $\perp$, separating these terminal classes from the dead unregistered class. The inherited preregistration classification is unchanged.*

The forward map sends each actual state to its region and the displayed key: its placement before registration, its retained label at a terminal, or its carrier, label and literal placement otherwise. For the inverse, choose an actual silent representative for $\star$ and an actual state for each registerable placement. Lemma 3.2 realizes every terminal label. For (C, L, P), Lemma 3.1 supplies a compatible complete chain at C (the retained chain itself when L is short); the inherited fixed-chain fibre theorem realizes that chain at P. Encoding any such representative recovers the chosen key. Conversely, decoding an encoded state may choose a different history, but preserves its region, label and placement where applicable. Equation (2.4) and the inherited future classification place it in the original observer class. Thus both composites are identities on the indicated quotient.

For fixed proper C, L, the placement fibre has size $n\binom{2N-r}{N-r}$. Summing over labels and carrier arcs gives (3.5). Lemma 3.2 gives the terminal count; the other regions contribute $R_n + 1$. Restricting to registered states removes those two regions.

For $r \geq 1$, the first two depths give

$$c_{r,1} = r(r+1), \qquad c_{r,2} = 2 + \frac{2}{3} r(r^2 - 1), \tag{3.7}$$

whereas $c_{0,j} = 1$ at every depth. The terminal counts are $T_{N,1} = 2N$ and $T_{N,2} = 2 + 2N(N-1)$.

Corollary 3.4.

Let s_n be the complete registered-summary count. For $j \geq N$,

$$Q_n^{[j]} = 1 + R_n + s_n - (n-1)\, 2\, 3^{N-1}. \tag{3.8}$$

Proof. *Every chain is now retained in full. There are $2\,3^{N-1}$ complete chains ending at checkpoint N. Each has n phases in a complete summary and one terminal prefix key here. All nonterminal placements remain literal. The only subtraction is therefore $n-1$ per full-carrier chain.*

Corollary 3.5.

For $1 \leq j < k$, the identity on actual states induces a surjection $X_n/{\equiv_k} \twoheadrightarrow X_n/{\equiv_j}$. Both the whole-domain and registered-domain counts strictly increase from depth j to $j+1$ whenever $1 \leq j < N$. Their equivalence relations stabilize for $j \geq N$.

Proof. *The depth-j label is recovered from the depth-k label by truncating the records and recomputing the short/saturated marker. The future-event contract is unchanged, so (2.4) gives $x \equiv_k y \Rightarrow x \equiv_j y$.*

For strictness, take the same first j singleton R absorptions, then absorb the entire remaining complement with final port binding L or R. Since $j < N$, the carrier before that last event is proper. The inherited chain realization theorem supplies both actual terminal states. Their depth-j labels agree and their futures are silent, but their depth-$(j+1)$ labels contain different complete final records. They therefore give two distinct finer classes inside one coarser class, in either state domain.

Every chain has length at most N. For $j \geq N$ its label determines the whole chain, and the marker is determined by that chain and j. Equality of labels consequently has the same meaning at every such depth. Equation (2.4) proves stabilization.

4 Fixed-Depth Asymptotics

Lemma 4.1.

For each fixed $j \geq 1$,

$$\sum_{r \geq 0} c_{r,j} 2^{-r} = 5\,2^j - 1. \tag{4.1}$$

Proof. *For $|x| < 1$ the two relevant sums are*

$$\sum_{r \geq 1} \binom{r+1}{j+1} x^r = \frac{x^j}{(1-x)^{j+2}}, \qquad \sum_{r \geq \ell} \binom{r-1}{\ell-1} x^r = \left(\frac{x}{1-x}\right)^{\ell}. \tag{4.2}$$

They follow by differentiating the geometric series, or by the ordinary generating function for compositions. At $x = 1/2$, the empty-chain contribution is 1, the saturated contribution is 2^{j+2}, and the short contributions sum to $2^j - 2$.

Theorem 4.2.

For fixed $j \geq 1$, as $n \to \infty$,

$$Q_n^{[j]} \sim \frac{5\,2^j}{16\sqrt{\pi}} \sqrt{n}\, 4^n, \qquad Q_{n,\mathrm{reg}}^{[j]} \sim \frac{5\,2^j - 1}{16\sqrt{\pi}} \sqrt{n}\, 4^n. \tag{4.3}$$

Proof. *Put $B_N = \binom{2N}{N}$. For $0 \leq r \leq N$,*

$$\frac{\binom{2N-r}{N-r}}{B_N} = \prod_{i=0}^{r-1} \frac{N-i}{2N-i} \leq 2^{-r}, \tag{4.4}$$

and for fixed r this tends to 2^{-r}. The coefficients $c_{r,j}$ have polynomial growth in r for fixed j. Extending the summand by zero for $r \geq N$, (4.4) gives a summable bound, so dominated convergence and Lemma 4.1 imply

$$\frac{A_n^{[j]}}{nB_N} \longrightarrow 5\,2^j - 1. \tag{4.5}$$

The terminal contribution is polynomial in N at fixed depth and is negligible on this scale. Moreover $R_n/(nB_N) = N/(N+1) \to 1$. Stirling's formula [7] gives the central-binomial asymptotic $B_N \sim 4^N/\sqrt{\pi N}$, so

$$nB_N \sim \frac{\sqrt{n}\, 4^n}{16\sqrt{\pi}}, \tag{4.6}$$

since $N = n - 2$. Combining the contributions proves (4.3).

The estimates below treat depths that grow with n.

5 A Uniform Truncated-Binomial Reduction

Set

$$S_{r,j} = \sum_{\ell=0}^{\min(j,r)} 2^{\ell} \binom{r}{\ell}, \qquad Z_{N,j} = \sum_{r=0}^{N-1} \binom{2N-r}{N-r} S_{r,j}. \tag{5.1}$$

Lemma 5.1.
For every $r \geq 0$ and $j \geq 1$,

$$\frac{S_{r,j}}{r+1} \leq c_{r,j} \leq (r+2) S_{r,j}. \tag{5.2}$$

Consequently

$$\frac{Z_{N,j}}{N} \leq \frac{A_n^{[j]}}{n} \leq (N+1) Z_{N,j}. \tag{5.3}$$

Proof. *At $r = 0$ both coefficients are one. For $r \geq 1$, the saturated term in (3.1) is at most $(r+1)2^j\binom{r}{j}$ when $j \leq r$, and is zero otherwise. Every short term is at most its corresponding term in $S_{r,j}$. This gives the upper bound.*

For $1 \leq \ell < j$ use $\binom{r-1}{\ell-1} = (\ell/r)\binom{r}{\ell}$. When $j \leq r$, also use $\binom{r+1}{j+1} \geq \binom{r}{j}$. Together these give $S_{r,j} - 1 \leq r c_{r,j}$, including $j > r$. There is at least one compatible label, so $c_{r,j} \geq 1$ and $S_{r,j} \leq (r+1)c_{r,j}$. Multiplying by the placement weight and using $r+1 \leq N$, $r+2 \leq N+1$ proves (5.3).

Lemma 5.2.
For all $N \geq 1$ and $j \geq 1$,

$$Z_{N,j} = \sum_{\ell=0}^{\min(j,N)} 2^{\ell} \left[\binom{2N+1}{N-\ell} - \binom{N}{\ell} \right]. \tag{5.4}$$

Proof. *Exchange the two finite sums in (5.1). For fixed ℓ, append the term $r = N$ and apply the convolution*

$$\sum_{r=\ell}^{N} \binom{r}{\ell} \binom{2N-r}{N} = \binom{2N+1}{N-\ell}. \tag{5.5}$$

To prove (5.5), choose an $(N+\ell+1)$-element subset of a $(2N+1)$-element ordered set, and let its $(\ell+1)$st member be at position $r+1$. Choose ℓ members before it and N after it. Complementing the chosen subset gives the right-hand binomial coefficient. The appended term is $\binom{N}{\ell}$.

Write $\mathcal{U}_{N,j} = \sum_{\ell \leq \min(j,N)} 2^{\ell}\binom{2N+1}{N-\ell}$. The subtracted sum in (5.4) is at most 3^N, whereas $\mathcal{U}_{N,j}$ contains $\binom{2N+1}{N} \geq B_N \geq 4^N/(2N+1)$. Therefore

$$0 \leq 1 - \frac{Z_{N,j}}{\mathcal{U}_{N,j}} \leq (2N+1)\left(\frac{3}{4}\right)^N = o(1) \tag{5.6}$$

uniformly in positive j. Also $T_{N,j} \leq 3^N$ and $R_n \leq nB_N$. Since $Z_{N,j} \geq B_N$, equations (3.6), (5.3) and (5.6) show that $\mathcal{U}_{N,j}$ determines the exponential rate of both observer counts, with only polynomial factors and an exponentially negligible subtraction.

6 The Linear-Memory Growth Spectrum

Let $H(x) = -x\log x - (1-x)\log(1-x)$, with $0\log 0 = 0$. We use the elementary bounds

$$\frac{e^{MH(k/M)}}{M+1} \le \binom{M}{k} \le e^{MH(k/M)} \qquad (0 \le k \le M). \tag{6.1}$$

For $0 < k < M$, take success probability k/M in the binomial distribution. Its kth probability is a mode, at least $1/(M+1)$ and at most one. Writing that probability explicitly gives (6.1). The endpoint cases are immediate.

Theorem 6.1.

Fix $0 < \theta < 1$ and put $j_n = \lfloor \theta n \rfloor$ on the cofinite range where $j_n \ge 1$. Then both limits exist, and

$$\lim_{n\to\infty} (Q_n^{[j_n]})^{1/n} = \lim_{n\to\infty} (Q_{n,\mathrm{reg}}^{[j_n]})^{1/n} = \lambda(\theta), \tag{6.2}$$

where λ is given by (1.1). In particular $4 < \lambda(\theta) < 9/2$ for $0 < \theta < 1/3$.

Proof. *With $N = n-2$ and $\ell/N \to \beta$, the rate of the summand in $\mathcal{U}_{N,j}$ is*

$$G(\beta) = \beta\log 2 + 2H\left(\frac{1-\beta}{2}\right) = (2+\beta)\log 2 - (1+\beta)\log(1+\beta) - (1-\beta)\log(1-\beta). \tag{6.3}$$

Indeed (6.1) bounds its logarithm by $\ell\log 2 + (2N+1)H((N-\ell)/(2N+1))$, with error at most $\log(2N+2)$. After division by N this expression converges uniformly to $G(\ell/N)$: entropy is uniformly continuous on $[0,1]$, and the change in its argument is uniformly $O(1/N)$. There are at most $N+1$ summands. The logarithm of their sum differs from that of the largest by at most $\log(N+1)$.

Since $j_n/N \to \theta$, compactness gives the upper rate $\max_{0\le\beta\le\theta} G(\beta)$. Conversely every $0 \le \beta < \theta$ can be approached by an admissible integer ℓ_N; continuity supplies the endpoint by approximation. Thus upper and lower rates agree. More explicitly at the threshold $\theta = 1/3$, $\ell_N = \lfloor N/3 \rfloor \le \lfloor n/3 \rfloor$ is admissible and $\ell_N/N \to 1/3$.

Differentiation in the interior gives

$$G'(\beta) = \log\frac{2(1-\beta)}{1+\beta}. \tag{6.4}$$

Hence G increases strictly from $G(0) = \log 4$ up to $\beta = 1/3$, where $G(1/3) = \log(9/2)$, and decreases after that point. The constrained maximum is at $\beta = \min(\theta, 1/3)$. Exponentiation gives (1.1), and the uniform comparison of Section 5 transfers it to both quotients. Finally $N/n \to 1$, so the same limit uses the exponent $1/n$.

Corollary 6.2.

If $j_n \ge 1$ eventually and $j_n = o(n)$, both observer counts have exponential base 4.

Proof. *In the same uniform estimate every admissible $0 \le \ell/N \le j_n/N$ tends uniformly to zero. The maximum rate is therefore $G(0) = \log 4$, and the central summand gives the matching lower bound.*

Above density $1/3$, the ratio to the complete-summary count also tends to one.

Lemma 6.3.

Let $h_{n,\ell}$ count complete registered summaries whose selected chain has length ℓ. For $0 \le \ell \le N$,

$$h_{n,\ell} = n\,2^\ell \binom{2N}{N-\ell}, \qquad s_n = n\sum_{\ell=0}^{N} 2^\ell \binom{2N}{N-\ell}. \tag{6.5}$$

Proof. *At $\ell = 0$ the carrier is F, and its placement fibre has size $n\binom{2N}{N}$. For $\ell \geq 1$, group complete chains by their last checkpoint r. There are $2^{\ell}\binom{r-1}{\ell-1}$ chains, each with $n\binom{2N-r}{N-r}$ placements. This includes $r = N$, where the fibre is the n full-carrier phases. The required convolution is*

$$\sum_{r=\ell}^{N} \binom{r-1}{\ell-1}\binom{2N-r}{N-r} = \binom{2N}{N-\ell}. \tag{6.6}$$

Choose an $(N+\ell)$-element subset of a $2N$-element ordered set and designate its ℓth member as position r. Choose $\ell - 1$ members before it and N after it. Complementation gives the right-hand side. The inherited actual chain/fibre classification therefore proves (6.5) for actual summaries.

There is a useful exact reweighting. If K has the auxiliary binomial distribution $\mathrm{Bin}(2N, 1/3)$, then

$$\frac{h_{n,\ell}}{n(9/2)^N} = \Pr(K = N - \ell), \qquad \frac{s_n}{n(9/2)^N} = \Pr(K \leq N). \tag{6.7}$$

Since

$$\Pr(K > N) \leq 2^{-N}\mathbb{E}[2^K] = 2^{-N}(4/3)^{2N} = (8/9)^N, \tag{6.8}$$

we obtain the explicit lower bound $s_n \geq n(9/2)^N[1 - (8/9)^N]$.

Corollary 6.4.
Fix $1/3 < \theta < 1$ and $j_n = \lfloor \theta n \rfloor$. Then

$$\frac{Q_n^{[j_n]}}{Q_n^{\mathrm{full}}} \longrightarrow 1, \qquad \frac{Q_{n,\mathrm{reg}}^{[j_n]}}{s_n} \longrightarrow 1. \tag{6.9}$$

More explicitly, fix any ρ with $1/3 < \rho < \theta$ and put $\delta_\rho = \log(9/2) - G(\rho) > 0$. For all sufficiently large n, each relative deficit in (6.9) is nonnegative and at most

$$\frac{(N+1)e^{-N\delta_\rho} + \frac{2}{3}(2/3)^N}{1 - (8/9)^N}. \tag{6.10}$$

Proof. *Let $Q_n^{[\infty]}$ denote the full-prefix quotient of Corollary 3.4. The map from full-summary classes to full-prefix classes forgets only terminal phase. Its exact loss is*

$$D_n = (n-1)\,2\,3^{N-1}. \tag{6.11}$$

Under the further map to depth-j classes, every full-prefix class with chain length at most j has a singleton fibre. Below depth j the short label recovers the whole chain. At length exactly j, the last retained record already gives the current carrier. A longer chain with that prefix would strictly enlarge the carrier, so it cannot have the same nonterminal placement; at a full carrier no further absorption is possible. Thus these classes do not merge with each other or with any longer-history class.

Set $L_{n,>j} = \sum_{\ell=j+1}^{N} h_{n,\ell}$, with an empty sum equal to zero. The number of remaining full-prefix classes is at most $L_{n,>j}$. Consequently

$$0 \leq Q_n^{\mathrm{full}} - Q_n^{[j]} \leq D_n + L_{n,>j}. \tag{6.12}$$

The same loss equals $s_n - Q_{n,\mathrm{reg}}^{[j]}$, because both whole-domain counts contain the identical $1 + R_n$ contribution.

Eventually $j_n \geq \rho N$. By (6.1), for $\beta = \ell/N$, $2^{\ell}\binom{2N}{N-\ell} \leq e^{NG(\beta)}$. The function G decreases strictly on $(1/3, 1]$, so

$$L_{n,>j_n} \le n(N+1)e^{NG(\rho)}. \tag{6.13}$$

Divide (6.12) by $Q_n^{\mathrm{full}} \ge s_n$, or by s_n for the registered ratio. Use (6.8) and $D_n/[n(9/2)^N] \le \frac{2}{3}(2/3)^N$ to get (6.10). Both terms in its numerator decay exponentially, proving (6.9).

For $0 < \theta < 1/3$, Theorem 6.1 and (6.7)–(6.8) instead give $Q_n^{[\lfloor \theta n \rfloor]}/Q_n^{\mathrm{full}} \to 0$. At the critical density, an exact count yields the transition between these ratios.

Theorem 6.5.

Fix $u \in \mathbb{R}$ and put

$$j_n = \left\lfloor \frac{n}{3} + u\sqrt{n} \right\rfloor.$$

For all sufficiently large n, $1 \le j_n \le N$. Then

$$\frac{Q_n^{[j_n]}}{Q_n^{\mathrm{full}}} \longrightarrow \Phi(3u/2), \qquad \frac{Q_{n,\mathrm{reg}}^{[j_n]}}{s_n} \longrightarrow \Phi(3u/2), \tag{6.14}$$

where Φ is the standard normal distribution function. In particular,

$$\frac{Q_n^{[\lfloor n/3 \rfloor]}}{Q_n^{\mathrm{full}}} \longrightarrow \frac{1}{2}. \tag{6.15}$$

Moreover, for every fixed $A > 0$, with $j_n(u) = \lfloor n/3 + u\sqrt{n} \rfloor$, both limits hold uniformly at Berry–Esseen rate; for all sufficiently large n, the indices satisfy $1 \le j_n(u) \le N$ simultaneously for $|u| \le A$:

$$\begin{aligned} \sup_{|u|\le A} \left| \frac{Q_n^{[j_n(u)]}}{Q_n^{\mathrm{full}}} - \Phi(3u/2) \right| &= O_A(n^{-1/2}), \\ \sup_{|u|\le A} \left| \frac{Q_{n,\mathrm{reg}}^{[j_n(u)]}}{s_n} - \Phi(3u/2) \right| &= O_A(n^{-1/2}). \end{aligned} \tag{6.16}$$

Proof. *Fix $A > 0$ and let $|u| \le A$ throughout. All estimates below are uniform over this interval. Write*

$$H_{<j} = \sum_{\ell=0}^{j-1} h_{n,\ell}, \qquad U_{N,j}^{\mathrm{short}} = \sum_{\ell=1}^{j-1} 2^\ell \binom{N-1}{\ell-1},$$

and

$$V_{N,j} = \binom{2N+1}{N-j} + \binom{2N+1}{N-j-1} - \binom{N}{j} - \binom{N}{j+1}.$$

Splitting the coefficient $c_{r,j}$ in (3.1) into its stopped and saturated-prefix terms, and applying (6.6), (5.5), and Pascal's identity, gives the exact identity for $1 \le j \le N$:

$$Q_{n,\mathrm{reg}}^{[j]} = H_{<j} - (n-1)U_{N,j}^{\mathrm{short}} + n2^j V_{N,j} + 2^j \binom{N}{j}. \tag{6.17}$$

Indeed, (6.6) sums the short-chain contributions over all complete summaries of lengths $1, \dots, j-1$. Removing the $r = N$ placements subtracts $nU_{N,j}^{\mathrm{short}}$, while the terminal labels restore $U_{N,j}^{\mathrm{short}}$; the net term is $H_{<j} - (n-1)U_{N,j}^{\mathrm{short}}$. For saturated prefixes, $\binom{r+1}{j+1} = \binom{r}{j+1} + \binom{r}{j}$ and (5.5) give the two first binomials in $V_{N,j}$. Removing $r = N$ gives its two negative binomials. The terminal saturated labels add $2^j\binom{N}{j}$.

In the window $j = j_n(u) = N/3 + u\sqrt{N} + O_A(1)$, put $x = (N-j)/(2N+1)$. Then $x = 1/3 - u/(2\sqrt{N}) + O_A(N^{-1})$. Since $H'(1/3) = \log 2$ and H'' is bounded near $1/3$, Taylor's formula gives

$$j \log 2 + (2N+1)H(x) = N \log(9/2) + O_A(1).$$

Uniform Stirling bounds in this interior window contribute the local $O(N^{-1/2})$ prefactor. Shifting the lower binomial index by one has the same bound. Thus

$$2^j \binom{2N+1}{N-j} + 2^j \binom{2N+1}{N-j-1} = O_A\left(\frac{(9/2)^N}{\sqrt{N}}\right).$$

Also $U^{\mathrm{short}}_{N,j} \le 2\,3^{N-1}$ and both endpoint terms $2^j\binom{N}{j}$ and $2^j\binom{N}{j+1}$ are at most 3^N. Because $s_n \ge n(9/2)^N[1-(8/9)^N]$ by (6.8), the short-chain and endpoint corrections in (6.17), after division by s_n, are $O((2/3)^N)$. The two local terms contribute $O_A(N^{-1/2})$, so

$$\frac{Q^{[j]}_{n,\mathrm{reg}}}{s_n} = \frac{H_{<j}}{s_n} + O_A(N^{-1/2}).$$

Moreover, (6.7)–(6.8) give $s_n/[n(9/2)^N] = 1 + O((8/9)^N)$. The whole-domain ratio differs from the registered-domain ratio by $O((R_n+1)/s_n)$, uniformly in u, because $0 \le Q^{[j]}_{n,\mathrm{reg}} \le s_n$. The exact formula $R_n = n\binom{2N}{N-1}$ and Stirling's estimate give $R_n/s_n = O(N^{-1/2}(8/9)^N)$.

Let $K \sim \mathrm{Bin}(2N, 1/3)$. By (6.7), $H_{<j}/s_n = \Pr(K > N-j \mid K \le N)$. The complement of the conditioning event has probability at most $(8/9)^N$ by (6.8), so conditioning changes this tail by an exponentially small amount. The Berry–Esseen inequality [1, 4] for the standardized binomial variable gives a uniform $O(N^{-1/2})$ error. Its threshold satisfies

$$\frac{N - j_n(u) - 2N/3}{(2/3)\sqrt{N}} = -\frac{3u}{2} + O_A(N^{-1/2}),$$

where the error includes the floor in $j_n(u)$ and $\sqrt{n} - \sqrt{N} = O(N^{-1/2})$. Since Φ is Lipschitz, this proves $H_{<j_n(u)}/s_n = \Phi(3u/2) + O_A(N^{-1/2})$. Combining with the prefix correction proves (6.14) and (6.16); setting $u = 0$ gives (6.15).

7 Computational Artifacts

The edition-specific evidence index under `experiments/paper44/` in the RIME repository links the Lean companion and finite controls. The companion proves the actual normal forms and both counts in Theorem 3.3 for all $n \ge 3$ and $j \ge 1$; its axiom audit records only `propext`, `Classical.choice` and `Quot.sound`.

Complete reachable-state enumeration and direct minimization agree with the normal-form partitions of Theorem 3.3 in all 20 cases with $3 \le n \le 7$ and $1 \le j \le n-1$.

n	Reachable states	Minimal counts at depths $1, \dots, n-1$
3	30	12, 12
4	176	69, 71, 71
5	920	402, 410, 414, 414
6	4,566	2,085, 2,283, 2,303, 2,311, 2,311
7	22,134	9,965, 11,803, 12,363, 12,411, 12,427, 12,427

Exact-arithmetic controls also check (6.5)–(6.8), (6.12) and (6.17) at finite sizes.

8 Claim Status and Boundary

Theorem 3.3 classifies the exact minimal quotient on reachable singleton-seed states for every positive prefix depth. Its finite structural and counting layer is covered by the separate Lean companion. The fixed-depth and linear-memory growth results and the critical-window estimate are manuscript proofs, not consequences of the finite controls or claims of Lean certification for the analytic layer.

The observer retains complete selected-event outputs and a prefix of past selected records, but not terminal physical phase. In particular, asymptotic equality of class counts in (6.9) neither identifies the quotient relations nor recovers every complete history.

9 Conclusion

The exact quotient separates the information needed for future events from the retained prefix of earlier records. On this fixed dynamics, varying prefix depth produces every exponential rate between 4 and 9/2, with a Gaussian transition in the critical window.

References

[1] Andrew C. Berry. The accuracy of the gaussian approximation to the sum of independent variates. *Transactions of the American Mathematical Society*, 49(1):122–136, 1941.

[2] WuJun Chen. Future-event minimal quotients for a circular registered absorption system, 2026. RIME Paper XLIII, version 1.0, Zenodo. DOI: 10.5281/zenodo.24148747.

[3] WuJun Chen. Registered absorption chains and placement fibers for a circular action with an adjacent idempotent, 2026. RIME Paper XLI, version 1.0, Zenodo. DOI: 10.5281/zenodo.24137725.

[4] Carl-Gustav Esseen. On the liapounoff limit of error in the theory of probability. *Arkiv för Matematik, Astronomi och Fysik*, 28A(9):1–19, 1942.

[5] George H. Mealy. A method for synthesizing sequential circuits. *The Bell System Technical Journal*, 34(5):1045–1079, 1955.

[6] Edward F. Moore. Gedanken-experiments on sequential machines. In Claude E. Shannon and John McCarthy, editors, *Automata Studies*, pages 129–154. Princeton University Press, 1956.

[7] National Institute of Standards and Technology. NIST Digital Library of Mathematical Functions, section 5.11: Asymptotic expansions. Stirling expansions for the gamma function; accessed 12 November 2026.

[8] Anil Nerode. Linear automaton transformations. *Proceedings of the American Mathematical Society*, 9(4):541–544, 1958.